Geometry of Ordinal Representations in Language Models

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Abstract

Recent work showed that language models represent character counts on curved 1D manifolds, with attention heads performing geometric transformations to enable computation. We test whether this generalizes across four ordinal tasks (bracket depth, indentation, table position, numeric magnitude) in Gemma-2-2B, Gemma-2-9B, and Qwen3-4B. We find that 1D manifolds with place-cell feature tiling emerge for tasks where the ordinal variable is locally computable from token identity, while tasks requiring cross-position integration or semantic extraction produce higher-dimensional or incoherent representations. Geometric computation is architecture-dependent: Qwen3-4B shows 4–10× stronger twisting than Gemma models for indentation, and its twisters preserve ordinal order ($\rho \geq 0.77$), unlike its numeric twisters ($\rho \leq 0.23$). Activation patching causally confirms that the manifold subspaces carry task-relevant information across all settings.

1. Introduction

Understanding how neural networks internally represent structured information is a central goal of mechanistic interpretability (Olah et al., 2020; Conmy et al., 2023). While much work has focused on identifying circuits and features (Elhage et al., 2022; Bricken et al., 2023; Templeton et al., 2024), recent studies have begun characterizing the *geometry* of learned representations.

Gurnee et al. (2026) demonstrated that Claude 3.5 Haiku represents character counts on curved 1D manifolds in the residual stream, discretized by sparse autoencoder (SAE) features that tile the manifold like place cells tile physical space (Moser et al., 2008). Attention heads perform geometric transformations (twisting and aligning the manifold) to compute distance to line boundaries. This raises a natural question: **do these geometric motifs generalize to other tasks requiring ordinal or counting representations, and do different model architectures learn the same geometry?**

We investigate four ordinal tasks across three open-weight models: Gemma-2-2B (Gemma Team, 2024), Gemma-2-9B (Gemma Team, 2024), and Qwen3-4B (Qwen Team, 2025). The tasks are bracket nesting depth, Python indentation level, markdown table position (row and column), and numeric magnitude. For each task and model, we (1) train linear probes to verify decodability, (2) discover manifold subspaces via PCA on per-value mean activations, (3) find SAE feature families using Gemma Scope (Lieberum et al., 2024) for the Gemma models, (4) analyze attention heads for geometric computation (twisting and alignment), and (5) causally validate manifold subspaces via activation patching.

Our main findings:

- **Manifold hypothesis partially generalizes.** Bracket depth and table column index produce 1D manifolds with place-cell feature tiling across all models. Indentation uses a 3D subspace. Numeric magnitude shows no coherent manifold despite strong linear decodability (train $R^2 > 0.93$).
- **Geometric computation is architecture-dependent.** Bracket depth manifolds are pre-organized from the embedding layer in all three models (max twist ≤ 0.13). Indentation and numeric magnitude show dramatically more twisting in Qwen3-4B (max 0.80 and 0.72) than in either Gemma model (≤ 0.33), despite similar probing accuracy.
- **Coherent vs. incoherent twisting.** High twist alone does not imply ordinal geometric computation. Qwen3-4B's indentation twisters preserve ordinal order ($\rho \geq 0.77$), while its numeric twisters do not ($\rho \leq 0.23$). This distinguishes heads that geometrically construct an ordinal manifold from heads that merely rearrange representations without structure.
- **Causal validation confirms manifold subspaces.** Activation patching of the top-k manifold directions causes large, specific drops in probe accuracy across

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all tasks and all three models, with negligible effect from random directions.

- **The key distinction is local computability.** Tasks where the ordinal variable is directly computable from token identity (brackets, table columns) produce clean 1D manifolds that require no geometric construction. Tasks requiring cross-position integration or semantic extraction produce higher-dimensional or incoherent representations.

2. Related Work

Representation geometry. The study of geometric structure in neural network representations has a rich history. Hewitt & Manning (2019) introduced structural probing to find syntax in word representations, establishing the methodology of projecting representations into task-relevant subspaces. The logit lens (nostalgebraist, 2020) and tuned lens (Belrose et al., 2023) revealed that intermediate representations progressively converge toward output predictions. Marks & Tegmark (2024) found emergent linear structure in LLM representations of truth values. Gurnee & Tegmark (2024) showed that LLMs represent continuous variables (space and time) as linear directions. Park et al. (2023) formalized the linear representation hypothesis, and Engels et al. (2025) extended it, showing that some features require multi-dimensional subspaces rather than single directions. Modell et al. (2026) argued that features are represented as manifolds and that cosine similarity encodes intrinsic manifold geometry. Shai et al. (2024) showed that transformers represent belief state geometry in their residual stream, including non-trivial fractal structures. Gurnee et al. (2026) made this concrete for counting tasks, showing that character counts trace curved 1D manifolds with SAE features acting as place-cell analogs. Our work directly extends this to multiple tasks and multiple open-weight models.

Theoretical grounding. Shai et al. (2026) proved that transformers learn factored representations in orthogonal subspaces, with dimensionality growing linearly when factors are conditionally independent. This predicts the orthogonal arrangement observed by Gurnee et al. (2026) and suggests it should generalize across tasks. Park et al. (2026) argued that representation geometry should be understood information-geometrically rather than in Euclidean terms, which is relevant for interpreting our curved manifold findings and for the causal steering experiments we perform.

Sparse autoencoders and features. Sparse autoencoders have emerged as a primary tool for decomposing neural network representations into interpretable features (Huben et al., 2024; Bricken et al., 2023). Templeton et al. (2024) scaled this approach to Claude 3 Sonnet, finding interpretable features at scale. Gemma Scope (Lieberum et al., 2024) provides pretrained SAEs for Gemma-2 models, which we use for feature family discovery. Michaud et al. (2025) studied how SAE scaling interacts with feature manifolds, finding that manifold structure affects the number of features needed; our results on numeric magnitude (high selectivity but zero monotonic features) are consistent with their finding that SAEs struggle to decompose manifold-structured features into clean families. Lubana et al. (2026) noted that SAEs assume temporal independence, potentially missing sequential structure in how manifolds evolve across positions, which is a relevant caveat for our ordinal tasks.

Mechanistic interpretability of structured tasks. Prior work has studied how transformers represent structured information including modular arithmetic (Nanda et al., 2023) and algorithmic tasks (Conmy et al., 2023). Beckmann & Queloz (2026) proposed a tiered framework for mechanistic understanding; if multiple tasks and multiple architectures use the same geometric motifs, this would constitute evidence for principled rather than task-specific or architecture-specific representations. Our cross-model comparison directly addresses this question.

Cross-architecture comparison. Gemma-2 (Gemma Team, 2024) uses grouped-query attention, SwiGLU activations, and alternating local/global attention. Qwen3-4B (Qwen Team, 2025) uses grouped-query attention, RoPE positional encoding, and SwiGLU activations, with a substantially larger head count (32 heads vs. 8 in Gemma-2-2B). These architectural differences, particularly the head count and positional encoding scheme, may explain the differences in geometric computation we observe.

3. Methods

3.1. Models and Tasks

We study three open-weight models: Gemma-2-2B (26 layers, 8 heads, $d = 2304$), Gemma-2-9B (42 layers, 16 heads, $d = 3584$), and Qwen3-4B (36 layers, 32 heads, $d = 2048$). Qwen3-4B uses grouped-query attention and RoPE positional encoding. We use Gemma Scope (Lieberum et al., 2024) residual stream SAEs (16K dictionary) for the Gemma models; no pretrained SAEs exist for Qwen3-4B.

We study four ordinal tasks. **Bracket nesting depth**: sequences of mixed brackets with per-token depth labels (0–10); 3,000 samples, half synthetic and half from The Stack. **Python indentation level**: per-token indent level (0–5 in units of 4 spaces); 3,000 samples from The Stack. **Markdown table position**: synthetic tables with per-token row (0–14) and column (0–8) indices; 2,000 tables. **Numeric magnitude**: text with numbers labeled by $\lfloor \log_{10}(\text{value}) \rfloor$

(0–8); 5,000 samples from OpenWebText. All sequences are tokenized with each model's native tokenizer (max length 256).

3.2. Probing, Manifold Discovery, and Feature Families

For each task, model, and layer ℓ, we collect residual stream activations at labeled token positions (4,000 train / 1,000 test). We train logistic regression probes (classification) or Ridge regression probes (numeric magnitude) and report test accuracy or R^2.

For manifold discovery, we compute the mean activation $\bar{\mathbf{h}}_v = \frac{1}{|S_v|}\sum_{i \in S_v} \mathbf{h}_i$ for each ordinal value v and apply PCA to the matrix of per-value means. We report the number of PCs needed to explain 90% of variance and pairwise cosine similarities between per-value means.

For the Gemma models, we use Gemma Scope SAEs at four evenly spaced layers to find feature families. For each feature j, the mean activation per ordinal value gives a tuning curve $\tau_j(v)$. We rank features by between-value variance, keeping those active on $> 5\%$ of tokens. A feature is *selective* if its peak exceeds $3\times$ its mean, and *monotonic* if its tuning curve is approximately non-decreasing or non-increasing.

3.3. Geometric Computation and Causal Validation

Following Gurnee et al. (2026), we measure how much each attention head rearranges the manifold by projecting per-value means into Q-space: $\bar{\mathbf{q}}_v^h = \mathbf{W}_Q^h \bar{\mathbf{h}}_v$. We compute three scores per head: the *twist score* $1 - \text{corr}(\mathbf{d}_{\text{resid}}, \mathbf{d}_Q)$ (high means the head rearranges pairwise distances); the *alignment score* (fraction of variance in PC1 of the Q-projected manifold); and the *ordinality score* $|\rho_{\text{Spearman}}(v, \text{PC1}(v))|$ (high means ordinal order is preserved).

To causally validate the manifold subspaces, we patch the top-k PCA directions of the per-value means by replacing a token's projection onto this subspace with that of a token with a different ordinal label, leaving the orthogonal complement unchanged. We compare the resulting probe accuracy drop against patching k random directions as a control. We report results at $k = 3$ for the layer with the largest manifold drop. We follow best practices from Zhang & Nanda (2024) and note the caveat of Makelov et al. (2024) that subspace patching can activate parallel pathways; our random-direction control is designed to mitigate this concern.

Table 1. Best test probing score (layer in parentheses) across models. Metric is accuracy except numeric (R^2). Manifold dimensionality is shown in Figure 4.

Task	Gemma-2-2B	Gemma-2-9B	Qwen3-4B
Brackets	0.538 (L2)	0.525 (L6)	0.573 (L2)
Table cols	0.983 (L12)	0.996 (L24)	0.992 (L16)
Indentation	0.943 (L12)	0.947 (L18)	0.944 (L24)
Table rows	0.746 (L16)	0.821 (L22)	0.803 (L22)
Numeric	0.536 (L16)	0.549 (L0)	0.688 (L2)

4. Results

4.1. Probing and Manifold Structure

Linear probes achieve high test accuracy for all tasks across all three models (Table 1). Indentation and table columns reach $\geq 94\%$ test accuracy by mid-network in every model. Bracket depth peaks at 53–57% test accuracy, which is low in absolute terms but well above chance (1/11 $\approx 9\%$), and with near-perfect *training* accuracy ($\geq 99\%$), indicating the representation is linearly separable but the probe generalizes imperfectly to held-out tokens. Table rows reach 74–82% test accuracy. Numeric magnitude achieves train $R^2 > 0.93$ in all models but test R^2 varies substantially: 0.54 for Gemma-2-2B, 0.55 for Gemma-2-9B at layer 0 (degrading to negative at deeper layers due to overfitting), and 0.69 for Qwen3-4B. The ordinal variable is linearly accessible in the residual stream for every task and model studied.

Manifold dimensionality. The tasks separate into consistent tiers across all three models:

1D manifolds. Bracket depth and table column index concentrate $> 90\%$ of variance in PC1 at mid-to-late layers (Figure 4). For brackets, PC1 explains 92–100% of variance across models (Gemma-2-2B: 96.5% at L8; Gemma-2-9B: 97.4% at L16; Qwen3-4B: 100% at L6–L20). For table columns, PC1 explains 90–100% across models. These replicate the curved 1D manifold found by Gurnee et al. (2026) for character counts. The cosine similarity heatmap for brackets (Figure 1) shows sharp block structure: depth 0 is in a distinct direction from depths 1+, with deeper values progressively more similar.

2–3D manifolds. Indentation requires 2–3 PCs for 90% variance across models (PC1 $\approx$ 47–54%). The six indent levels are spread across multiple dimensions, likely because indent level interacts with code structure (function depth, class nesting). Table rows require 1–4 PCs depending on model and layer, with Qwen3-4B collapsing to a 1D manifold at mid-layers (PC1 $= 99.5\%$ at L6) while Gemma models use 3–4D.

4–5D manifolds. Numeric magnitude requires 4–5 PCs across all models, with PC1 explaining only 40–62% of variance. The 3D PCA projection shows no clear ordinal progression; values are scattered without systematic arrangement, consistent with the model encoding magnitude via multiple overlapping representations rather than a single ordinal manifold. The existence of multi-dimensional manifolds for indentation (3D) and numeric magnitude (4–5D) is consistent with Engels et al. (2025), who showed that some LLM features require multi-dimensional subspaces.

4.2. Feature Families: Place Cells for Some Tasks

We analyze SAE feature families for Gemma-2-2B and Gemma-2-9B (Table 2).

Brackets (best case). At layer 8 in Gemma-2-2B, 14/20 top features are selective and 2/20 are strictly monotonic (Figure 2). Feature #9213 is a sharp depth=0 detector (activation 83.6 at depth 0, <1 at depth 2+). Feature #302 increases monotonically from 5.7 to 16.9 across depths. Feature #13542 peaks at depth 10. In Gemma-2-9B, 7–9/20 features are selective at layers 4–14, with similar place-cell structure (e.g., feature #2967 at L14: monotonically increasing from 6.0 to 17.8).

Indentation (weak). Only 2–8/20 features are selective across layers and models. Feature #8675 at L2 in Gemma-2-2B is a clean indent=4 detector (tuning curve: [0.1, 0.4, 0.2, 0.0, 4.7, 0.0]), but most features are broadly tuned. Selectivity degrades at deeper layers (2/20 at L20 in Gemma-2-2B; 2–3/20 at L24–34 in Gemma-2-9B).

Table rows (moderate). 5–10/20 features are selective. Most top features peak at row 0 (start of table). Feature #9768 at L20 in Gemma-2-2B is a clean monotonically decreasing row detector (25.0 $\rightarrow$ 0.0). The SAE preferentially encodes “start of table” over fine-grained row position.

Numeric magnitude (no tiling). Despite 6–12/20 features being selective, *zero* features are monotonic at any layer in either model. Tuning curves are erratic (e.g., #15567 at L14 in Gemma-2-2B: [24.0, 6.6, 4.2, 8.2, 1.1, 2.1, 18.1, 1.2, 0.0]). The SAE cannot decompose numeric magnitude into an orderly feature family. This contrasts with Heinzerling & Inui (2024), who found monotonic representations of numeric properties in encoder models; the difference may reflect the autoregressive training objective or the log-scale binning we use. The absence of clean tiling is consistent with Dehaene (2003)’s logarithmic mental number line: a log-scale representation compresses large numbers, making fine-grained ordinal distinctions harder to encode as discrete features.

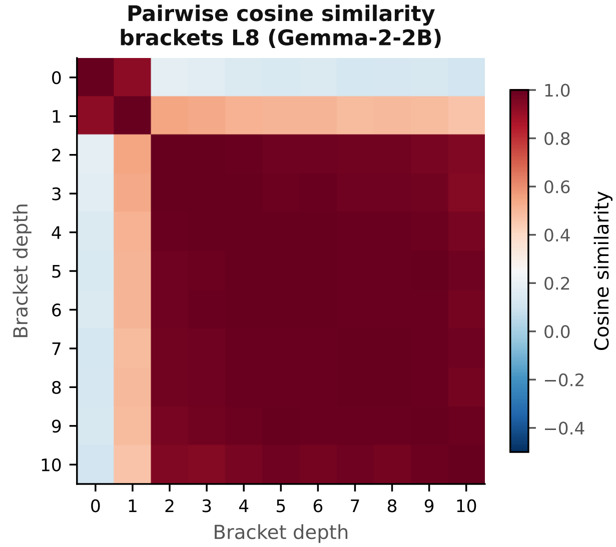

Figure 1. **Pairwise cosine similarity of bracket depth mean activations** (Gemma-2-2B, L8). Depth 0 occupies a distinct direction; depths 2–10 are nearly identical, confirming a 1D manifold with one outlier at depth 0.

4.3. Geometric Computation Is Architecture-Dependent

We analyzed all attention heads per model per task (Table 3). The key finding is that the degree of geometric computation varies substantially across architectures, even when probing accuracy is similar.

Brackets: pre-organized across all models. All three models show low maximum twist scores (0.033, 0.060, 0.131). Residual alignment starts high (0.65–0.78 at layer 0) and climbs to near-perfect (0.98–1.00) by mid-network. Nearly all heads achieve high ordinality ($\rho \geq 0.97$) in Q-space without rearranging the manifold. The twist heatmap (Figure 3, left) shows uniformly low activity (max 0.033), confirming no head performs geometric rearrangement. This contrasts with Gurnee et al. (2026), where specific heads performed dramatic twists for character counting.

Indentation: architecture-dependent twisting. The Gemma models show moderate, diffuse twisting (max 0.21–

Table 2. SAE feature family quality: selective and monotonic features out of top 20, at the layer with highest selectivity. Gemma models only (no pretrained SAEs for Qwen3-4B).

Task	Gemma-2-2B		Gemma-2-9B	
	Sel.	Mono.	Sel.	Mono.
Brackets	15/20 (L14)	4/20	9/20 (L4)	5/20
Indentation	7/20 (L8)	4/20	8/20 (L14)	4/20
Table rows	8/20 (L8)	2/20	10/20 (L4)	1/20
Numeric	12/20 (L2)	0/20	10/20 (L4)	0/20

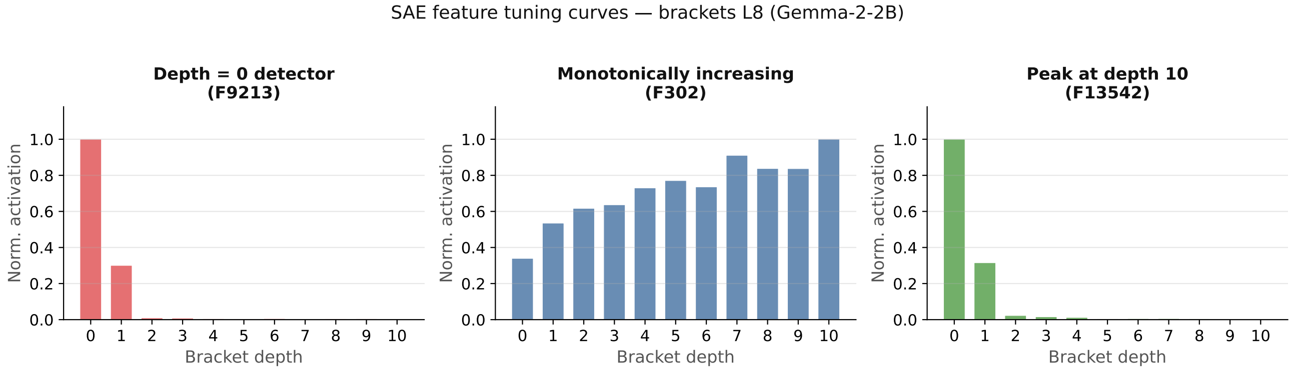

Figure 2. **SAE feature tuning curves — brackets L8 (Gemma-2-2B).** Three archetypes of place-cell tiling: a sharp depth=0 detector (F9213), a monotonically increasing feature (F302), and a feature peaking at maximum depth (F13542). Activations normalized to $[0, 1]$ per feature.

Table 3. Geometric computation summary. Max twist_q score and residual alignment range (min→max across layers). G-2-2B/9B = Gemma-2-2B/9B.

Task	Model	Max twist	Resid. align.
Brackets	G-2-2B	0.033	0.78→0.98
	G-2-9B	0.060	0.72→0.98
	Qwen3-4B	0.131	0.65→1.00
Indentation	G-2-2B	0.214	0.47→0.64
	G-2-9B	0.331	0.43→0.52
	Qwen3-4B	0.803	0.46→0.98
Table rows	G-2-2B	0.215	0.44→0.74
	G-2-9B	0.135	0.62→0.87
	Qwen3-4B	0.237	0.51→1.00
Numeric	G-2-2B	0.210	0.32→0.50
	G-2-9B	0.288	0.32→0.52
	Qwen3-4B	0.724	0.37→0.95

0.33, mean 0.036–0.040) with residual alignment stuck at 0.43–0.64. Qwen3-4B shows dramatically stronger twisting: max 0.803 (L12H11), mean 0.160, with residual alignment climbing from 0.46 to 0.98 (Figure 3, right; Figure 4). The top Qwen3-4B twisters (L12H11: 0.803, L9H9: 0.728, L13H2: 0.723) all maintain high ordinality ($\rho \geq 0.77$), meaning they rearrange the manifold while preserving ordinal order, which is exactly the behavior Gurnee et al. (2026) found for character counting. This suggests Qwen3-4B performs active geometric construction of the indentation manifold, while Gemma models do not.

Table rows: moderate twisting, consistent across models. All three models show moderate maximum twist (0.135–0.237) concentrated in early-to-mid layers. The top twisters are L10H0 (Gemma-2-2B, 0.215), L18H8 (Gemma-2-9B, 0.135, ord_q$= 1.0$), and L3H12 (Qwen3-4B, 0.237). Residual alignment climbs substantially in all models (0.44→0.74 in Gemma-2-2B; 0.51→1.00 in Qwen3-4B), indicating active manifold organization.

Numeric: high twisting in Qwen3-4B, moderate in Gemma. Gemma models show moderate, diffuse twisting (max 0.21–0.29) with low residual alignment (0.32–0.52) throughout; the manifold never organizes. Qwen3-4B shows high twisting (max 0.724, mean 0.286) with residual alignment climbing to 0.95, but the top twisters have low ordinality (L13H24: ord_q$= 0.23$), meaning the head rearranges the manifold without preserving ordinal order. This is incoherent geometric computation rather than the ordinal-preserving twisting seen for indentation.

4.4. Causal Validation and Cross-Task Analysis

Activation patching confirms that the manifold subspaces causally carry task-relevant information (Table 4). Patching the top-3 manifold directions causes large drops in probe accuracy (0.26–0.83 for classification tasks; >8 R^2 units for numeric magnitude), while patching 3 random directions causes negligible drops (<0.01 in all cases except Qwen3-4B numeric). The ratio of manifold drop to random drop exceeds 28× for every task–model combination.

For brackets, despite low test accuracy (53–57%), patching the manifold subspace still causes drops of 0.23–0.45 across models and layers (Table 4), confirming the representation is geometrically organized even though the probe overfits to training tokens.

Notably, patching 3 directions of a 1D manifold (brackets, table cols) causes drops comparable to patching 3 directions of a 5D manifold (numeric). This indicates that even high-dimensional representations concentrate their task-relevant information in a small subspace, though for numeric magnitude that subspace lacks the ordinal structure seen in the 1D tasks.

The numeric magnitude results deserve special attention.

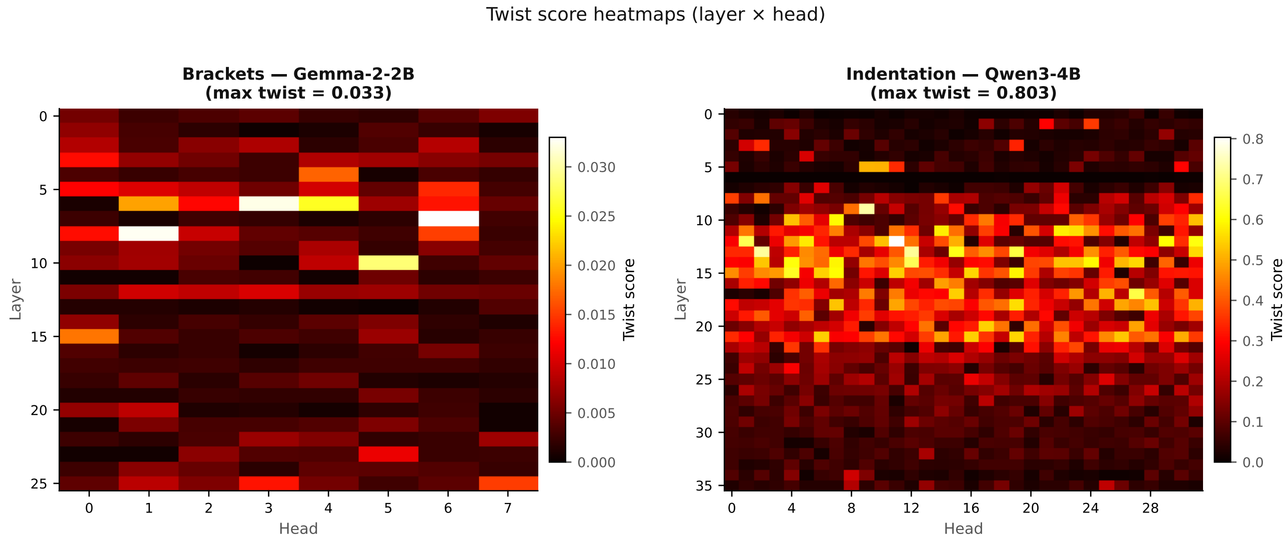

Figure 3. **Twist score heatmaps (layer × head).** *Left*: Brackets in Gemma-2-2B (max twist = 0.033) — activity is low and diffuse, confirming the manifold is pre-organized and no head performs geometric rearrangement. *Right*: Indentation in Qwen3-4B (max twist = 0.803) — bright activity concentrated in layers 5–20, with multiple heads performing strong ordinal-preserving twists. Note the 25× difference in scale between panels.

The manifold drop is enormous (>8 R^2 units) despite the representation lacking clean manifold structure. This indicates that even a diffuse, high-dimensional representation can be causally concentrated in a few directions. For Qwen3-4B, the random drop is also elevated (3.10), suggesting the numeric representation is less geometrically organized and more sensitive to arbitrary perturbations.

Cross-task and cross-model feature sharing. Features #9213, #15567, #7214, and #6631 appear among the top-20 features for multiple tasks in Gemma-2-2B. However, all four are primarily depth/value=0 detectors that fire strongly at the start of structured content across tasks. These are better characterized as “beginning of structure” features than general ordinal position detectors.

Cross-task overlap of top-20 twisting heads is modest and model-dependent. In Gemma-2-2B, numeric ∩ table_rows shares 8 heads, and brackets ∩ table_rows shares 6. In Gemma-2-9B and Qwen3-4B, most pairwise overlaps are 0–3 heads. No head appears in the top-20 twisters for all four tasks in any model, and no head is consistently a top twister across all three models.

5. Discussion

When do manifolds emerge? Our results suggest a taxonomy based on how the ordinal variable relates to token identity. Tasks where the variable is *locally computable* from token identity (bracket depth depends on whether the current token is an opener/closer; table column index on pipe characters) produce clean 1D manifolds that are pre-organized from the embedding layer. This holds consistently across all three models. Tasks requiring *cross-position integration* (table row index requires counting newlines; indentation requires tracking block structure) produce higher-dimensional manifolds (2–4D). Tasks requiring *semantic extraction* (numeric magnitude depends on parsing multi-digit numbers and context) produce diffuse, high-dimensional representations (4–5D) without clean manifold structure.

This distinction between decodability and geometric structure is important: a linear probe succeeding (train $R^2 > 0.93$ for numeric magnitude) does not imply the representation has manifold structure. The information may be distributed across many dimensions without the low-dimensional, curved geometry that characterizes the manifold regime.

Architecture-dependent geometric computation. The most striking finding is that the degree of geometric computation varies dramatically across architectures, even when probing accuracy is similar. Qwen3-4B shows 4–10× higher maximum twist scores than the Gemma models for indentation (0.803 vs. 0.214–0.331) and numeric magnitude (0.724 vs. 0.210–0.288). For indentation, Qwen3-4B’s top twisters maintain high ordinality ($\rho \geq 0.77$), suggesting genuine ordinal-preserving geometric computation. For numeric magnitude, the top twisters have low ordinality, suggesting incoherent rearrangement rather than structured computation.

Importantly, the twist concentration ratio (max/mean) is similar across models for indentation: Gemma-2-2B ≈ 5.9,

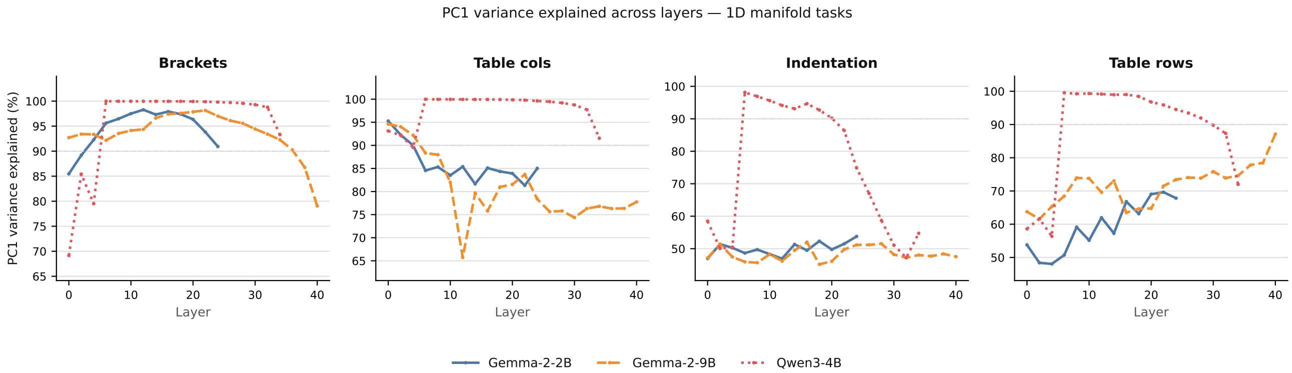

Figure 4. **PC1 variance explained across layers.** For brackets and table columns, PC1 captures $> 90\%$ of variance (dashed line) in all three models, confirming 1D manifold structure. For indentation and table rows, Qwen3-4B collapses to a 1D manifold at mid-layers (step-jump to $\sim$100%) while Gemma models remain at 45–65%, revealing architecture-dependent manifold organization.

Table 4. Activation patching results (top-3 manifold directions, best layer). Man. drop: decrease in test metric when patching manifold subspace. Rand. drop: decrease when patching random directions.

Task	Model	Baseline	Man. drop	Rand. drop
Brackets	Gemma-2-2B	0.496	0.342	0.002
	Gemma-2-9B	0.456	0.264	−0.004
	Qwen3-4B	0.464	0.415	0.001
Indentation	Gemma-2-2B	0.926	0.492	−0.001
	Gemma-2-9B	0.947	0.665	0.001
	Qwen3-4B	0.901	0.519	−0.001
Table rows	Gemma-2-2B	0.718	0.583	0.009
	Gemma-2-9B	0.819	0.690	0.001
	Qwen3-4B	0.744	0.633	0.010
Table cols	Gemma-2-2B	0.919	0.738	<0.001
	Gemma-2-9B	0.942	0.831	<0.001
	Qwen3-4B	0.859	0.648	0.007
Numeric	Gemma-2-2B	0.508	8.24	0.001
	Gemma-2-9B	−2.92†	20.0	0.025
	Qwen3-4B	0.243	22.2	3.10

† Gemma-2-9B numeric probe overfits severely (train $R^2 = 0.99$, test $R^2 = -2.92$). The large manifold drop (20.0) reflects that patching disrupts the representation, but the absolute value should be interpreted cautiously given the broken baseline.

Gemma-2-9B ≈ 8.3, Qwen3-4B ≈ 5.0. This means the difference is not that Qwen3-4B concentrates twisting in fewer specialized heads; rather, all its heads twist more strongly in absolute terms. This argues against the "more heads = more specialization" hypothesis and instead suggests that Qwen3-4B's architecture (RoPE positional encoding, different attention pattern) leads to a qualitatively different computational strategy for constructing ordinal manifolds.

Why no twisting for brackets? The absence of twisting heads for bracket depth is consistent across all three models. We hypothesize that bracket depth is so directly encoded in token identity (opening brackets increment, closing brackets decrement) that the embedding layer already places tokens at the correct manifold position. The model needs only to *read* the manifold, not *construct* it. This contrasts with character counting (Gurnee et al., 2026), where the count depends on all preceding characters and must be computed incrementally.

Limitations. The bracket depth test accuracy (53–57%) is low despite near-perfect training accuracy, suggesting the probe overfits to training tokens. The numeric magnitude results for Gemma-2-9B show strongly negative test R^2 at deeper layers, indicating severe overfitting of the Ridge regression probe. The synthetic bracket data has a minor labeling inconsistency (always closing with ')' regardless of opener type), affecting only the synthetic half of the dataset. No pretrained SAEs exist for Qwen3-4B, limiting our feature family analysis to the Gemma models. The twist score metric operates on per-value *mean* activations and may miss geometric transformations that operate on individual token representations.

Future directions. Training SAEs on Qwen3-4B would test whether the stronger geometric computation we observe corresponds to cleaner feature families. The information-geometric framework of Park et al. (2026) may better characterize curved manifolds than our Euclidean PCA approach, particularly for the higher-dimensional cases. Extending to larger models (Gemma-2-27B, Qwen3-14B) would test whether the architecture-dependent differences we observe scale with model size or are specific to the 2–4B parameter range.

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